Hidden Losses in Interrupted Markets: Evidence from the 2025 CME Trading Halt

Fernando Méndez Benítez

November 2025

Abstract

Following the 2025 CME trading outage, we develop an estimation framework to quantify the "hidden losses" associated with the interruption by treating the outage as a temporary illiquidity shock. We evaluate a suite of Discount for Lack of Marketability (DLOM) models across a basket of futures contracts and assess their performance against synthetic outage scenarios during the preceding week. Identifying the optimal framework, we apply a linear adjustment calibrated to the synthetic regime to produce a final estimate of hidden losses incurred during the observed trading halt.

1 Background

The unexpected trading halt at CME Group on 28 November 2025 created a rare natural experiment, disrupting trading activity across major futures contracts and exposing market participants to an abrupt and system-wide liquidity interruption. Although option-theoretic Discount for Lack of Marketability ("DLOM") models are extensively documented in the context of restricted securities, their application to exchange-level trading halts has received comparatively little attention, largely due to the infrequency of such events. Nevertheless, their conceptual relevance to temporary non-tradability is well recognised in the literature [4].

DLOM frameworks seek to quantify the expected adverse effect faced by an investor who is unable to transact for a finite period of time. In the context of a trading halt, this precisely corresponds to determining the discount required for a halted asset relative to a hypothetical continuously tradable counterpart to compensate for the loss of market access during the interruption.

Chaffe [1] proposes valuing this loss using an at-the-money European put, which captures strictly downside movements: any dollar decline in the underlying translates one-for-one into intrinsic value. While this approach captures adverse price movements, it does not account for forgone upside.

Longstaff [4], by contrast, models the investor as possessing perfect timing ability in the absence of restrictions and prices the DLOM using a floating-strike lookback put. The resulting discount reflects the expected maximum price over the restriction window. Although theoretically consistent, this framework generates explosive discount levels under high volatility or long horizons, often exceeding 100%.

Finnerty [2] and Ghaidarov [3] argue that perfect timing is implausible and instead advocate a floating-strike Asian put, modelling an investor who is equally likely to transact at any point during the restriction period. Despite its intuitive positioning as an intermediate case, the dependence between spot and path-average often results in discounts that fall below those of the Chaffe model. Moreover, Finnerty's revised model is bounded above, with no discount exceeding 32.3%.

In earlier work [5], we introduced a hybrid model designed to capture both immediate downside risk and a portion of forgone upside without the explosive behaviour observed in the Longstaff specification. Its payoff is defined as the maximum of an at-the-money put and a floating-strike Asian put, thereby retaining full exposure to day-zero adverse moves and incorporating a realistic measure of upside potential. A closed-form valuation and numerical validation of the hybrid option were previously established.

Because DLOM values are risk-neutral expectations of unobservable quantities, empirical validation typically relies on restricted stock studies, block trades, or pre-IPO transitions. To obtain a more grounded estimate relevant to the 2025 CME halt, we instead calibrate the models directly to observed synthetic outage outcomes during the surrounding trading regime.

2 Methodology

To quantify the relationship between model-implied and realised adverse effects, we compute for each day of the week preceding 28 November 2025:

Observed Losses: the adverse effect obtained from the realised payoff of each of the Chaffe, Ghaidarov, Longstaff, and Hybrid specifications at the end of day.

Modelled Losses: the analytic value of the corresponding option at the start of day, using the day's opening price and contemporaneous implied volatility.

This procedure is applied across a representative basket of futures contracts: Natural Gas, Crude Oil, Gold, Palladium, NASDAQ 100 E-mini, S&P 500 E-mini, and Bitcoin. For each framework, we compute the root-mean-square error (RMSE) between modelled and observed losses across the four pre-halt days. Restricting the calibration to this specific regime mitigates distortions from volatility clustering, drift, or structural breaks.

Upon identifying the optimal-performing model, we fit a linear regression of realised losses on their theoretical counterparts to obtain a scalar adjustment factor. While such proportionality is unlikely to hold globally, it is plausible over short horizons and within a coherent volatility regime. Using the regression-adjusted option values and outbreak-day implied volatilities, we compute estimated adverse effects for each asset during the halt. Multiplying these values by the corresponding open interest yields indicative estimates of total portfolio losses.

3 Results

We first present representative synthetic outage results for Gold and Bitcoin futures.

Table 1: Synthetic Outage Gold Loss%

Nov. Date	24	25	26	27
Ghaidarov (Model)	0.24	0.24	0.25	0.26
Ghaidarov (Realised)	0.00	0.00	0.22	0.00
Chaffe (Model)	0.40	0.41	0.42	0.45
Chaffe (Realised)	0.00	0.00	0.16	0.00
Hybrid (Model)	0.24	0.24	0.25	0.26
Hybrid (Realised)	0.00	0.00	0.22	0.00
Longstaff (Model)	0.82	0.82	0.86	0.91
Longstaff (Realised)	0.02	0.15	0.48	0.05

Table 2: Synthetic Outage BTC Loss%

Nov. Date	24	25	26	27
Ghaidarov (Model)	0.45	0.62	0.64	0.77
Ghaidarov (Realised)	0.00	0.05	0.00	0.16
Chaffe (Model)	0.77	1.07	1.10	1.32
Chaffe (Realised)	0.00	0.36	0.00	0.44
Hybrid (Model)	0.45	0.62	0.64	0.77
Hybrid (Realised)	0.00	0.36	0.00	0.44
Longstaff (Model)	1.57	2.18	2.24	2.68
Longstaff (Realised)	0.12	0.90	0.36	0.96

The pre-halt period was broadly bullish, causing most realised losses to fall well below their option-theoretic counterparts. This reinforces the need to evaluate performance exclusively within the prevailing regime.

The overall RMSE and the regression slope for each framework appear in the table below.

Table 3: Overall Model Performance

Framework	RMSE	Slope
Ghaidarov	0.42	0.14
Chaffe	0.71	0.14
Hybrid	0.40	0.26
Longstaff	1.31	-0.75

The Hybrid model produces the lowest RMSE and exhibits superior performance across all assets individually. Using the calibrated scalar adjustment, we estimate outage-day losses for each asset as follows.

Table 4: Estimated Losses During Trading Halt

Underlying	Open Interest (Contracts)	Open Interest (USD)	Loss (%)	Loss (USD)
Gold	7,712	32,249,270	0.06	20,495
Palladium	454	663,521	0.14	908
Nat. Gas	246,702	1,152,345	0.19	2199
Crude Oil	365,350	21,493,541	0.09	19,385
NASDAQ	297,716	7,556,627,512	0.06	4,372,513
S&P	2,001,126	13,685,200,433	0.04	5,655,321
BTC	23,183	2,142,456,945	0.19	3,982,575

Per-contract losses range from 0.04% to 0.19%, yet the aggregate portfolio impact totals approximately $14 million. While based on an illustrative basket, these results highlight the economic significance of even short-lived halts, particularly for high-volatility underlyings.

4 Concluding Remarks

Ultimately, the 2025 CME trading halt offered a rare opportunity to examine the behaviour of option-theoretic marketability models under an acute, system-wide interruption. By treating the outage as an illiquidity shock and benchmarking several established DLOM frameworks against synthetic outage scenarios in the days immediately preceding the event, we were able to evaluate not only their relative accuracy but also their practical suitability for estimating the adverse effects borne by investors during the halt.

Across assets and regimes, the results highlight the importance of restricting the evaluation window to the prevailing volatility and directional regime. The lead-up week exhibited predominantly upward drift, resulting in systematically higher model-implied discounts relative to realised losses. Within this context, the Hybrid Framework consistently delivered the lowest RMSE, outperforming both the downside-only Chaffe and the upside-maximising Longstaff approaches, while avoiding the persistent underestimation observed in Finnerty–Ghaidarov-type models. The subsequent linear adjustment, though reliant on short-horizon proportionality between modelled and observed effects, provided a stable and intuitive mapping from theoretical discount to realised impact.

Applying this calibrated hybrid measure to the outage-day open interest yields indicative hidden losses of roughly \$14 million for the representative portfolio and \$10 million across the NASDAQ and S&P mini futures, with individual instruments exhibiting markedly different sensitivities depending on their volatility and liquidity profiles. Although modest in percentage terms, these losses underscore the materiality of even brief market interruptions, particularly in highly leveraged or volatility-prone products.

More broadly, these findings illustrate the value of embedding marketability based methods within a regime-specific, empirically anchored framework when assessing the consequences of trading halts. While the rarity of such events precludes universal generalisation, the methodology presented here provides a tractable, internally consistent mechanism for quantifying short-term illiquidity costs in real time.

5 Appendix I – Data

Table 5: Daily Basket Precious Metals Data

Underlying	Nov.	Open	Close	Average	Max	ATM Vol.
Gold	24	4060.5	4088.7	4079.6	4089.5	19.6
Gold	25	4132.0	4160.0	4152.3	4166.3	19.7
Gold	26	4160.1	4151.1	4157.7	4171.0	20.4
Gold	27	4144.3	4187.4	4157.4	4189.5	21.7
Gold	28	4181.7	-	-	-	20.3
Palladium	24	1389.5	1389.5	1389.5	1389.5	48.4
Palladium	25	1400.0	1408.5	1404.6	1410.0	47.5
Palladium	26	1410.5	1427.0	1414.2	1430.5	44.9
Palladium	27	1421.0	1421.0	1421.0	1421.0	48.0
Palladium	28	1461.5	-	-	-	43.6

Table 6: Daily Basket Energy Commodities Data

Underlying	Nov.	Open	Close	Average	Max	ATM Vol.
Nat. Gas	24	4.51	4.53	4.52	4.53	65.4
Nat. Gas	25	4.48	4.49	4.49	4.50	62.2
Nat. Gas	26	4.50	4.61	4.57	4.64	62.0
Nat. Gas	27	4.62	4.65	4.60	4.66	63.1
Nat. Gas	28	4.67	-	-	-	60.8
Crude Oil	24	57.74	58.61	58.27	58.61	30.3
Crude Oil	25	58.08	58.20	58.12	58.23	34.4
Crude Oil	26	58.18	58.35	58.08	58.71	31.6
Crude Oil	27	58.36	59.03	58.75	59.03	33.5
Crude Oil	28	58.83	-	-	-	28.7

Table 7: Daily Basket Equity Index Data

Underlying	Nov.	Open	Close	Average	Max	ATM Vol.
NASDAQ	24	24392.8	24936.5	24735.5	24947.5	22.4
NASDAQ	25	25082.3	25170.5	25156.5	25189.5	20.7
NASDAQ	26	25173.0	25324.5	25240.0	25371.8	19.2
NASDAQ	27	25333.3	25346.3	25312.1	25346.3	19.6
NASDAQ	28	25382.0	-	-	-	18.4
S&P	24	6628.8	6725.5	6691.0	6727.3	16.6
S&P	25	6782.8	6797.8	6795.5	6801.3	14.5
S&P	26	6798.3	6829.8	6814.6	6845.5	13.4
S&P	27	6832.5	6833.8	6828.7	6837.3	13.7
S&P	28	6838.8	-	-	-	13.2

Table 8: Daily Basket Digital Asset Data

Underlying	Nov.	Open	Close	Average	Max	ATM Vol.
BTC	24	86015.0	88275.0	87189.0	88380.0	37.3
BTC	25	87365.0	87320.0	87635.4	88105.0	51.7
BTC	26	87220.0	91175.0	87999.1	91490.0	53.1
BTC	27	91045.0	90900.0	91296.6	91770.0	63.5
BTC	28	92415.0	-	-	-	59.2

References

[1] Chaffe, D.B.H. *Option Pricing as a Proxy for Discount for Lack of Marketability in Private Company Valuations*. Business Valuation Review, 1993.

[2] Finnerty, J.D. *The Impact of Transfer Restrictions on Stock Prices*. SSRN Electronic Journal, 2002.

[3] Ghaidarov, S. *Analysis and Critique of the Average Strike Put Option Marketability Discount Model*. SSRN Electronic Journal, 2009.

[4] Longstaff, F.A. *How Much Can Marketability Affect Security Values?* The Journal of Finance, 1995.

[5] Méndez Benítez, F. *A Hybrid Framework for the Discount for Lack of Marketability*. Working Paper, 2025.